

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	29 September 2026
Team ID	SWTID-2026-2603
Project Name	AnswerBot – AI-Powered FAQ & Customer Support Automation
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed AnswerBot solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through email/password form; input validation; secure password storage using bcrypt.
FR-2	User Authentication	Login using email and password; JWT-based authentication for protected API access.
FR-3	FAQ Management	Create, view, update and delete FAQ records; validate required fields.
FR-4	FAQ Search	Keyword/semantic-oriented search of FAQ content and category-based filtering.
FR-5	Category Management	Create, update and organize FAQ categories for structured support content.
FR-6	AI FAQ Generation	Send topic/question to Google Gemini and receive an FAQ draft for user review.
FR-7	Role Management	Apply access control for Admin, Content Creator, Authenticated User and Public User functions.
FR-8	Error Handling	Return centralized, understandable errors for invalid requests, authentication failures and unavailable services.

Non-functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide a simple workflow for login, search, FAQ management and AI-assisted generation.
NFR-2	Security	Use JWT authentication, bcrypt password hashing, protected routes, schema validation and role-based access.
NFR-3	Reliability	Handle database/API errors centrally and avoid saving invalid FAQ records.
NFR-4	Performance	Provide efficient FAQ search and lightweight REST API processing; avoid unnecessary database operations.
NFR-5	Availability	Keep the API, database and AI integration available according to the selected deployment environment; failures should return clear errors.
NFR-6	Scalability	Use a modular REST API and MongoDB/Mongoose structure that can support additional FAQ categories, users and features.

Requirement Summary

The solution combines secure authentication, centralized FAQ CRUD operations, search, category management and Gemini-assisted FAQ generation. AI-generated content remains subject to user review before being stored and reused.